

Basic Statistical Methods

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Configuring R

Functions from these packages will be used throughout this document:

```

library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t

```

Here are some R settings I use in this document:

```

rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE

```

Acknowledgements

This content is adapted from Vittinghoff et al. (2012), Chapter 3.

1 Introduction

This appendix reviews fundamental statistical methods that are prerequisites for the main content of this course. Most of this material should be familiar from Epi 202 and Epi 203.

Example dataset: HERS

Throughout this appendix we use the HERS dataset as a running example.

The “heart and estrogen/progestin study” (HERS) was a clinical trial of hormone therapy for prevention of recurrent heart attacks and death among 2,763 post-menopausal women with existing coronary heart disease (CHD) (Hulley et al. 1998).

The trial was conducted at 20 US clinical centers. Participants were randomized to receive either conjugated equine estrogens (0.625 mg/day) plus medroxyprogesterone acetate (2.5 mg/day) or a matching placebo (Hulley et al. 1998). Women were followed for an average of 4.1 years (Hulley et al. 1998).

The primary outcome was nonfatal myocardial infarction or CHD death (Hulley et al. 1998).

```
library(haven)
hers <- haven::read_dta(
  paste0(
    "https://regression.ucsf.edu/sites/g/files",
    "/tkssra6706/f/wysiwyg/home/data/hersdata.dta"
  )
)
```

2 Descriptive Statistics

See Vittinghoff et al. (2012), §3.2.

2.1 Summary statistics for continuous variables

Definition 2.1 (Sample mean). The **sample mean** of n observations $x_1, \dots, x_n$ is:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Definition 2.2 (Sample variance). The **sample variance** is:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

The divisor $n-1$ (rather than n) makes s^2 an unbiased^a estimator of the population variance σ^2 .

^a[estimation.qmd#def-unbiased](#)

Definition 2.3 (Sample standard deviation). The **sample standard deviation** is $s = \sqrt{s^2}$. It is expressed in the same units as the original data, making it more interpretable than the variance.

Definition 2.4 (Sample median). The **sample median** is the middle value when observations are sorted in ascending order. For n observations:

- If n is odd, the median is the $\frac{n+1}{2}$ th order statistic.
 - If n is even, the median is the average of the $\frac{n}{2}$ th and $\frac{n}{2} + 1$ th order statistics.
- The median is more robust to outliers than the mean.

Definition 2.5 (Interquartile range). The **interquartile range (IQR)** is the difference between the 75th percentile (the third quartile, Q_3) and the 25th percentile (the first quartile, Q_1):

$$\text{IQR} = Q_3 - Q_1$$

Like the median, the IQR is robust to outliers.

2.2 Summary statistics for categorical variables

Definition 2.6 (Sample proportion). For a binary outcome, the **sample proportion** of “successes” (coded as 1) is:

$$\hat{p} = \frac{k}{n}$$

where k is the number of successes and n is the total sample size.

2.3 Computing summary statistics in R

The `tbl_summary()` function from the `gtsummary` package produces formatted summary tables:

Example 2.1 (HERS baseline summary statistics).

```

library(dplyr)
library(gtsummary)
library(haven)

hers |>
  mutate(
    HT = haven::as_factor(HT),
    exercise = haven::as_factor(exercise),
    smoking = haven::as_factor(smoking),
    diabetes = haven::as_factor(diabetes)
  ) |>
  select(age, BMI, glucose, SBP, DBP, HT, exercise, smoking, diabetes) |>
  tbl_summary(
    by = HT,
    statistic = list(
      gtsummary::all_continuous() ~ "{mean} ({sd})",
      gtsummary::all_categorical() ~ "{n} ({p}%)"
    ),
    digits = gtsummary::all_continuous() ~ 1,
    label = list(
      age ~ "Age (years)",
      BMI ~ "BMI (kg/m²)",
      glucose ~ "Fasting glucose (mg/dL)",
      SBP ~ "Systolic BP (mmHg)",
      DBP ~ "Diastolic BP (mmHg)",
      exercise ~ "Exercises regularly",
      smoking ~ "Current smoker",
      diabetes ~ "Diabetes"
    )
  ) |>
  add_overall() |>
  bold_labels()

```

Table 1

Characteristic	Overall N = 2,763 ¹	placebo N = 1,383 ¹	hormone therapy N =
Age (years)	66.6 (6.7)	66.8 (6.7)	66.5 (6.6)
BMI (kg/m²)	28.6 (5.5)	28.5 (5.5)	28.6 (5.5)
Unknown	5	4	1
Fasting glucose (mg/dL)	112.2 (36.8)	112.4 (36.8)	111.9 (36.9)
Systolic BP (mmHg)	135.1 (19.0)	135.1 (19.4)	135.0 (18.7)
Diastolic BP (mmHg)	73.2 (9.7)	73.1 (9.7)	73.2 (9.7)
Unknown	1	1	0
Exercises regularly	1,068 (39%)	530 (38%)	538 (39%)
Current smoker	360 (13%)	182 (13%)	178 (13%)
Diabetes	731 (26%)	352 (25%)	379 (27%)

¹Mean (SD); n (%)

2.4 Exploratory data analysis

Graphical summaries reveal aspects of the data distribution that summary statistics may miss, such as skewness, multimodality, and outliers.

2.4.1 Histograms

A **histogram** displays the distribution of a continuous variable by dividing the range of values into intervals (bins) and plotting the number or proportion of observations in each bin.

Example 2.2 (Distribution of fasting glucose).

```
library(ggplot2)

hers |>
  ggplot() +
  aes(x = glucose) +
  geom_histogram(binwidth = 5, fill = "steelblue", color = "white") +
  labs(
    x = "Fasting glucose (mg/dL)",
    y = "Count"
  )
```

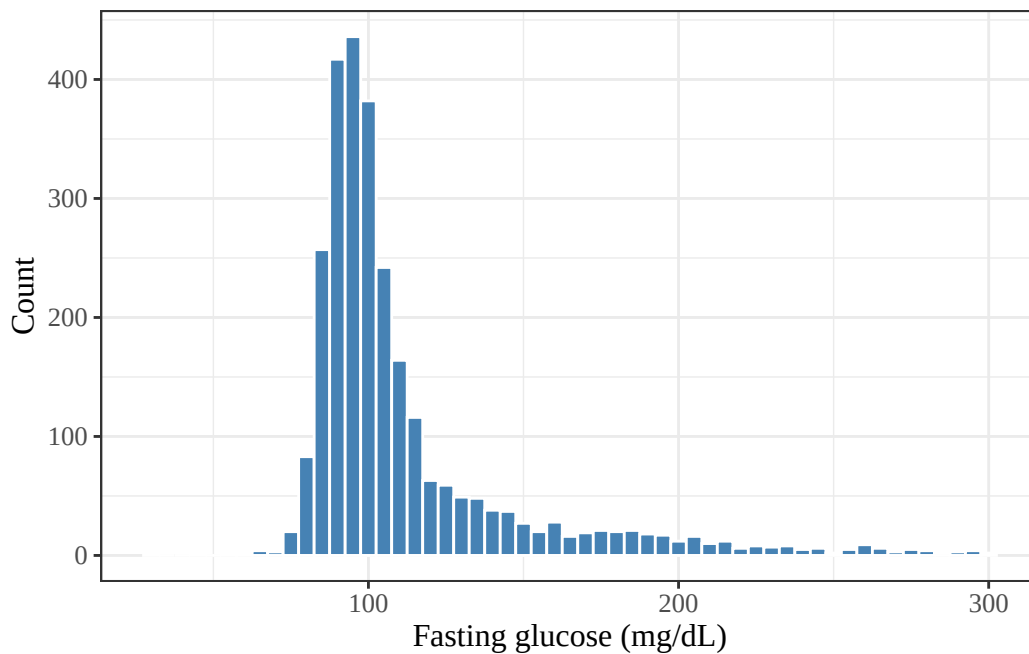


Figure 1: Distribution of fasting glucose (mg/dL) in HERS participants

2.4.2 Boxplots

A **boxplot** (box-and-whisker plot) summarizes the distribution of a continuous variable using five statistics: the minimum, the first quartile (Q_1), the median, the third quartile (Q_3), and the maximum (with outliers plotted separately).

Example 2.3 (Fasting glucose by hormone therapy group).

```

hers |>
  mutate(HT = haven::as_factor(HT)) |>
  ggplot() +
  aes(x = HT, y = glucose) +
  geom_boxplot() +
  labs(
    x = "Hormone therapy",
    y = "Fasting glucose (mg/dL)"
  )

```

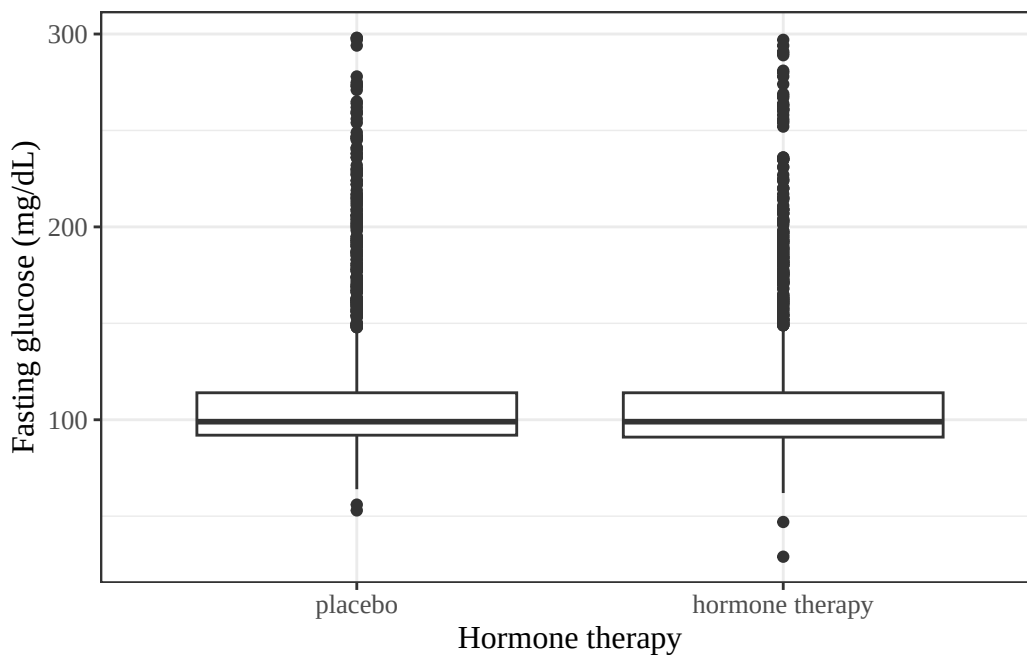


Figure 2: Fasting glucose by hormone therapy assignment in HERS

2.4.3 Scatterplots

A **scatterplot** displays the joint distribution of two continuous variables by plotting each observation as a point.

Example 2.4 (BMI versus fasting glucose).

```

hers |>
  ggplot() +
  aes(x = BMI, y = glucose) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    x = "BMI (kg/m2)",
    y = "Fasting glucose (mg/dL)"
  )

```

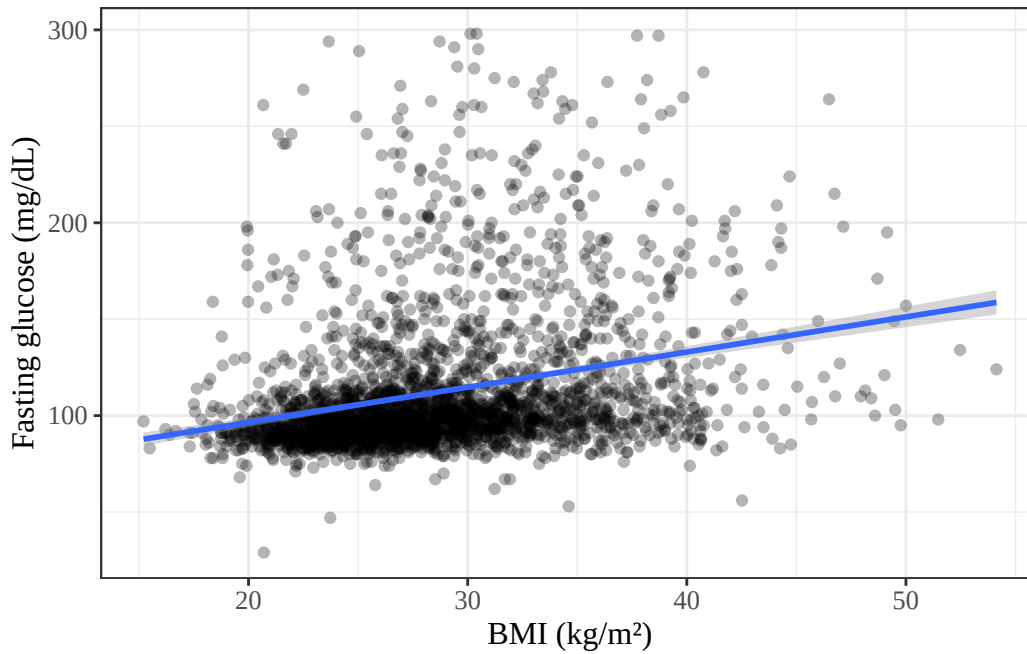


Figure 3: Fasting glucose versus BMI in HERS participants

3 Comparing Two Groups: Continuous Outcomes

See Vittinghoff et al. (2012), §3.3.

3.1 Hypotheses

Definition 3.1 (Null hypothesis). The **null hypothesis** H_0 is a specific claim about the population parameter(s) that we test against the data. In a two-group comparison of means, the null hypothesis is typically that the two group means are equal:

$$H_0 : \mu_1 = \mu_2$$

Definition 3.2 (Alternative hypothesis). The **alternative hypothesis** H_1 (or H_A) is the claim we are trying to find evidence for. For a two-sided test:

$$H_1 : \mu_1 \neq \mu_2$$

3.2 The two-sample t-test

Definition 3.3 (Two-sample t-test). The **two-sample t-test** (Welch's t-test) tests whether the means of two independent groups are equal.

For samples of sizes n_1 and n_2 from two groups with sample means $\bar{x}_1$, $\bar{x}_2$ and sample variances s_1^2 , s_2^2 , the test statistic is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Under H_0 , this statistic follows (approximately) a t -distribution with degrees of freedom

estimated by the Welch-Satterthwaite equation.

i Welch's t-test vs. pooled t-test

Welch's t-test (the default in R's `t.test()`) does not assume equal variances across groups. The pooled t-test assumes equal variances ($\sigma_1^2 = \sigma_2^2$) and pools the two sample variances into a single estimate. Welch's t-test is generally preferred because the equal-variance assumption is rarely verifiable in practice (Vittinghoff et al. 2012, sec. 3.3).

Example 3.1 (Comparing fasting glucose between hormone therapy groups). We test $H_0 : \mu_{\text{placebo}} = \mu_{\text{HT}}$ vs. $H_1 : \mu_{\text{placebo}} \neq \mu_{\text{HT}}$.

```
glucose_placebo <- hers |> filter(HT == 0) |> pull(glucose)
glucose_HT      <- hers |> filter(HT == 1) |> pull(glucose)

t.test(glucose_HT, glucose_placebo)
#>
#> Welch Two Sample t-test
#>
#> data: glucose_HT and glucose_placebo
#> t = -0.4246, df = 2761, p-value = 0.671
#> alternative hypothesis: true difference in means is not equal to 0
#> 95 percent confidence interval:
#> -3.34503 2.15423
#> sample estimates:
#> mean of x mean of y
#> 111.854 112.449
```

3.3 One-sample t-test

Definition 3.4 (One-sample t-test). The **one-sample t-test** tests whether the mean of a single population equals a specified null value μ_0 :

$$H_0 : \mu = \mu_0 \quad H_1 : \mu \neq \mu_0$$

The test statistic is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Under H_0 , $t \sim t_{n-1}$ (a t-distribution with $n - 1$ degrees of freedom).

3.4 Paired t-test

Definition 3.5 (Paired t-test). The **paired t-test** compares two related measurements (e.g., pre- and post-treatment values from the same subjects). Let $d_i = x_{i,1} - x_{i,2}$ be the within-subject difference; the test reduces to a one-sample t-test on the differences:

$$H_0 : \mu_d = 0 \quad H_1 : \mu_d \neq 0$$

Example 3.2 (Change in glucose over follow-up).

```
# glucose1 is follow-up glucose; glucose is baseline
t.test(hers$glucose1, hers$glucose, paired = TRUE)
#>
#> Paired t-test
#>
#> data: hers$glucose1 and hers$glucose
#> t = 4.151, df = 2612, p-value = 3.42e-05
#> alternative hypothesis: true mean difference is not equal to 0
#> 95 percent confidence interval:
#> 1.38248 3.85824
#> sample estimates:
#> mean difference
#> 2.62036
```

3.5 Confidence intervals for the difference in means

A two-sided $(1 - \alpha) \cdot 100\%$ confidence interval for $\mu_1 - \mu_2$ is:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{df}^* \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where t_{df}^* is the appropriate critical value from the t-distribution. The confidence interval is returned by `t.test()` in R alongside the hypothesis test result.

For more on confidence intervals, see Statistical Inference¹.

4 One-Way Analysis of Variance

Analysis of variance (ANOVA) generalizes the two-sample t-test to compare means across $k \geq 2$ groups.

Definition 4.1 (One-way ANOVA). In a **one-way ANOVA**, we test:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

vs. the alternative that at least one mean differs.

The F-statistic compares the **between-group variance** to the **within-group variance**:

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}} = \frac{SS_{\text{between}}/(k-1)}{SS_{\text{within}}/(n-k)}$$

Under H_0 , $F \sim F_{k-1, n-k}$.

Example 4.1 (Fasting glucose by race/ethnicity).

```
aov_result <- aov(glucose ~ factor(raceth), data = hers)
summary(aov_result)
#>
#>          Df Sum Sq Mean Sq F value Pr(>F)
#> factor(raceth)    2  45919   22959    17.1 4.1e-08 ***
#> Residuals      2760 3704543    1342
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

¹[inference.qmd#sec-CI](#)

i ANOVA as a special case of linear regression

One-way ANOVA is equivalent to a linear regression model with a single categorical predictor. See Linear Models Overview^a for details.

^a[Linear-models-overview.qmd](#)

5 Comparing Two Groups: Categorical Outcomes

See Vittinghoff et al. (2012), §3.5.

5.1 Contingency tables

Definition 5.1 (Contingency table). A **contingency table** (cross-tabulation) displays the joint frequencies of two categorical variables. For two binary variables, this is a 2×2 table with cells a, b, c, d :

	Outcome = 1	Outcome = 0	Total
Exposure = 1	a	b	$a + b$
Exposure = 0	c	d	$c + d$
Total	$a + c$	$b + d$	n

Example 5.1 (Exercise by hormone therapy group).

```
hers |>
  mutate(
    HT = haven::as_factor(HT),
    exercise = haven::as_factor(exercise)
  ) |>
  gtsummary::tbl_cross(
    row = exercise,
    col = HT,
    label = list(
      exercise ~ "Exercises regularly",
      HT ~ "Hormone therapy"
    ),
    percent = "row"
  )
```

Table 2: Exercise by hormone therapy group in HERS

	Hormone therapy		Total
	placebo	hormone therapy	
Exercises regularly			
no	853 (50%)	842 (50%)	1,695 (100%)
yes	530 (50%)	538 (50%)	1,068 (100%)
Total	1,383 (50%)	1,380 (50%)	2,763 (100%)

5.2 The chi-square test

Definition 5.2 (Pearson chi-square test). The **Pearson chi-square test** tests whether two categorical variables are independent. For a 2×2 table, the test statistic is:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where O_{ij} is the observed cell count and $E_{ij} = \frac{(\text{row total}) \times (\text{column total})}{n}$ is the expected cell count under independence.

Under H_0 (independence), $\chi^2 \sim \chi_1^2$ for a 2×2 table.

Example 5.2 (Chi-square test: exercise vs. hormone therapy).

```
chisq.test(hers$exercise, hers$HT)
#>
#> Pearson's Chi-squared test with Yates' continuity correction
#>
#> data:  hers$exercise and hers$HT
#> X-squared = 0.1016, df = 1, p-value = 0.75
```

5.3 Fisher's exact test

Definition 5.3 (Fisher's exact test). **Fisher's exact test** computes the exact probability of observing a 2×2 table at least as extreme as the observed table, given the marginal totals and under the null hypothesis of independence.

It is preferred over the chi-square test when cell counts are small (typically when any expected cell count is less than 5).

Example 5.3 (Fisher's exact test).

```
fisher.test(hers$exercise, hers$HT)
#>
#> Fisher's Exact Test for Count Data
#>
#> data:  hers$exercise and hers$HT
#> p-value = 0.725
#> alternative hypothesis: true odds ratio is not equal to 1
#> 95 percent confidence interval:
#>  0.879664 1.202192
#> sample estimates:
#> odds ratio
#>  1.02836
```

5.4 Measures of association for 2×2 tables

See Odds Ratios and Relative Risks² for definitions and formulas.

6 Correlation

See Vittinghoff et al. (2012), §3.6.

²[OR-RR.qmd](#)

6.1 Pearson correlation coefficient

Definition 6.1 (Pearson correlation coefficient). The **Pearson correlation coefficient** measures the strength and direction of the linear association between two continuous variables X and Y :

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

r ranges from -1 (perfect negative linear relationship) to $+1$ (perfect positive linear relationship); $r = 0$ indicates no linear association.

Example 6.1 (Correlation between BMI and glucose).

```
cor.test(hers$BMI, hers$glucose, method = "pearson")
#>
#> Pearson's product-moment correlation
#>
#> data:  hers$BMI and hers$glucose
#> t = 14.88, df = 2756, p-value <2e-16
#> alternative hypothesis: true correlation is not equal to 0
#> 95 percent confidence interval:
#>  0.237760 0.306865
#> sample estimates:
#>      cor
#> 0.272664
```

6.2 Spearman rank correlation

Definition 6.2 (Spearman rank correlation). The **Spearman rank correlation** r_S is the Pearson correlation computed on the *ranks* of the observations. It measures the strength and direction of any *monotone* association (not just linear) and is more robust to outliers.

Example 6.2 (Spearman correlation between BMI and glucose).

```
cor.test(hers$BMI, hers$glucose, method = "spearman")
#>
#> Spearman's rank correlation rho
#>
#> data:  hers$BMI and hers$glucose
#> S = 2.33e+09, p-value <2e-16
#> alternative hypothesis: true rho is not equal to 0
#> sample estimates:
#>      rho
#> 0.333751
```

7 Simple Linear Regression

See Vittinghoff et al. (2012), §3.6 and Linear Models Overview³.

³[Linear-models-overview.qmd](#)

7.1 Model specification

Definition 7.1 (Simple linear regression model). A **simple linear regression** model relates a continuous outcome Y to a single predictor X :

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

- β_0 is the **intercept**: the expected value of Y when $X = 0$.
- β_1 is the **slope**: the expected change in Y per one-unit increase in X .
- ε_i are independent Gaussian errors with mean 0 and variance σ^2 .

7.2 Ordinary least squares estimation

The parameters β_0 and β_1 are estimated by minimizing the **residual sum of squares (RSS)**:

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

The closed-form **ordinary least squares (OLS)** estimators are:

$$\hat{\beta}_1 = r \cdot \frac{s_y}{s_x}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where r is the Pearson correlation and s_x , s_y are the sample standard deviations.

7.3 Fitting a simple linear regression in R

Example 7.1 (Glucose on BMI).

Table 3

```
slr_fit <- lm(glucose ~ BMI, data = hers)
summary(slr_fit)
#>
#> Call:
#> lm(formula = glucose ~ BMI, data = hers)
#>
#> Residuals:
#>      Min       1Q   Median       3Q      Max
#> -81.55 -18.98 -10.35   3.76 190.81
#>
#> Coefficients:
#>              Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   60.074      3.565    16.9   <2e-16 ***
#> BMI           1.822      0.122    14.9   <2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 35.5 on 2756 degrees of freedom
#> (5 observations deleted due to missingness)
#> Multiple R-squared:  0.0743, Adjusted R-squared:  0.074
#> F-statistic: 221 on 1 and 2756 DF, p-value: <2e-16
```

The estimated slope is $\hat{\beta}_1 = 1.82$ mg/dL per kg/m², meaning fasting glucose increases by approximately 1.82 mg/dL for each 1 kg/m² increase in BMI.

7.4 The coefficient of determination (R^2)

Definition 7.2 (Coefficient of determination). The **coefficient of determination** R^2 measures the proportion of the total variance in Y that is explained by the linear regression on X :

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

R^2 ranges from 0 (no linear relationship) to 1 (perfect linear fit). For simple linear regression, $R^2 = r^2$.

7.5 Further reading

For a more thorough treatment of linear regression, see Linear Models Overview⁴ and Vittinghoff et al. (2012), Chapters 4 and 9.

Hulley, Stephen, Deborah Grady, Trudy Bush, et al. 1998. “Randomized Trial of Estrogen Plus Progestin for Secondary Prevention of Coronary Heart Disease in Postmenopausal Women.” *JAMA : The Journal of the American Medical Association* (Chicago, IL) 280 (7): 605–13.

Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. <https://doi.org/10.1007/978-1-4614-1353-0>.

⁴[Linear-models-overview.qmd](#)